

Extraction and isolation methods for lignin separation from sugarcane bagasse: A review

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ABSTRACT

This paper intends to review different extraction and isolation methods recently explored for lignin separation from sugarcane (*Saccharum officinarum* L.) bagasse. Initial bagasse processing is a pre-treatment mechanical action (milling) that facilitates applying further conversion technologies. There are several methods that have been developed for the extraction and isolation of the solubilized lignin from sugarcane bagasse; these include alkaline and ionic liquid methods. The insoluble lignin can be separated by the treatment of concentrated sulfuric acid. The quantity of sugarcane lignin fraction depends on the techniques applied and the aim of studies, but it has a range of 17–32% of biomass while the average value is about 23% of biomass. This fraction value has a major content in non-wood biomass and it is rich with aromatic chemical compounds.

1. Introduction

The lignin (*lignum*) means wood and it is a component of lignocellulose which consists also of cellulose and hemicellulose. This material is a natural complex biopolymer structure of carbohydrates.

An increased interest has been noted recently in the extraction of lignin from non-wood biomass such as, sugarcane, wheat (*Triticum aestivum*) straw, bamboo (*Bambusa vulgaris*), alfalfa (*Medicago sativa*), kenaf (*Hibiscus cannabinus*) and flax (*Linum usitatissimum*) fiber (Río et al., 2004; Ververis et al., 2004; Zhang et al., 2013). The focus of this paper is concentrated only on the sugarcane bagasse as biomass raw material of lignin source. This waste fiber is considered as the amount of remaining material after milling in sugar manufacturing or as residues in-land after harvests. It consists of two types of herbaceous plant fibers: rind and pith, which are longer than hardwood fibers and shorter than softwood fibers (Arni and Converti, 2012). All factories that are producing sugar from sugarcane have amount residues of this waste fiber, and typically produced about 30% of bagasse. It contains about 10–15% sucrose and 12–16% fiber (Arni and Converti, 2012). This biomaterial has a certain value, especially because of its sustainable and its economic interest for production fine chemical from renewable resource.

Today, there are a lot of bagasse applications; a search in literature data shows a number of studies have been done on sugarcane bagasse use and its application (Sene et al., 2001; Arni, 2004a,b; Zilli et al., 2004; Junior et al., 2006; Doherty et al., 2007; Boussarsar et al., 2009; Griffin, 2011; Cunha et al., 2011; Rocha et al., 2015a,b; Wahba et al., 2015; Costa et al., 2017). There are different levels for the reuse of bagasse as a raw material starting from domestic to commercial use. It

is used to cultivate edible mushrooms, such as shiitake (*Lentinula edodes* (Berk.)) (Arni and Converti, 2012), also it is used as fertilizers (Espinoza-Acosta et al., 2014). The sugarcane bagasse is used in industry to produce power as a fuel to generate electricity (Deepchand, 2005), to make paper (Poopak and Reza, 2012), in building materials (Loh et al., 2013) and feedstock to different products based on fermentation for the production of bio-based materials (Carvalho et al., 2002a,b; Carvalho et al., 2004; Carvalho et al., 2005; Converti et al., 2003; Santos et al., 2003; Santos et al., 2005a,b; Moldes et al., 2006; Sarrouh et al., 2007; Cunha et al., 2006,2009; Capecchia et al., 2015). Bagasse also used as biosorbents for the removal of heavy metals ions such as Cd, Cu, Pb, Zn, Ni and Cr (VI) ions (Junior et al., 2006), also for oil absorbing in the absence of water (Said et al., 2009). The summary of usage of bagasse is presented in Fig. 1.

The main focus of the present paper is the delignification processes that employed in the treatment of sugarcane bagasse to extract and isolate lignin fraction.

2. Processes for decomposition of lignocellulosic materials

Technologies used for bagasse decomposition are the same used in agricultural waste biomass decomposition. Usually, the decomposition processes depend on the ending goal of treatment, biomass type and quantity. Many researchers have established various processing techniques that can be used to convert these materials into various new products (Sun et al., 2000; Lavarack et al., 2002; Ou et al., 2009; Laopaiboon et al., 2010; Li and Ge, 2011; Rocha et al., 2011, 2012, 2015a).

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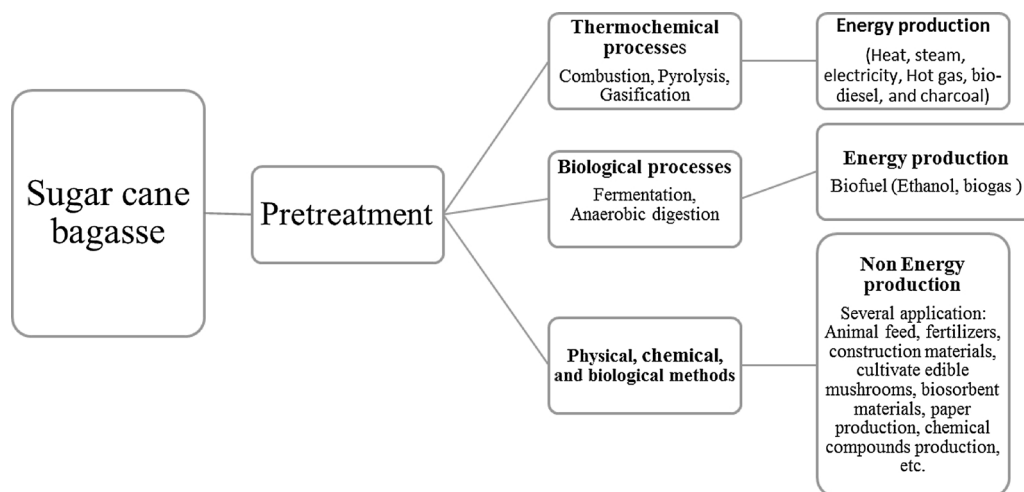


Fig. 1. Schematic diagram of processing and potential uses of bagasse.

There are two general categories of technologies for conversion of bagasse, firstly, the processing for conversion into energy products and secondly, processing for conversion into non energy products. Fig. 1 gives a general overview of processes and potential uses of sugarcane bagasse.

Processing of energy products aims to produce power in its different forms, such as, heat and steam, electricity, synthetic fuel oil, charcoal, producer gas, methane, ethanol, methanol and bio-diesel. The main conversion processes to energy products are thermochemical processes, such as, combustion, pyrolysis, and gasification. The most frequent using of sugarcane bagasse for energy production is the direct combustion (process based on solid-state) which consumed on-site (Mullinger and Jenkins, 2008; Arni et al., 2010c).

The biological processes to convert bagasse into energy are based on fermentation processes to produce ethanol, and Anaerobic Digestion (AD) to produce biogas (Converti and Del Borghi, 1996; Converti, 1998; Converti et al., 1998, 1999). Saxena et al. (2009) presented a review article on various biochemical processes for conversion of biomass into biological hydrogen gas and ethanol.

Biological pretreatment is sometimes used in combination with chemical treatment to solubilize lignin in order to make cellulose more accessible to hydrolysis and fermentation. Different microorganisms are able to partially or completely degrade biomass which is described elsewhere (Pandey et al., 2000). In addition, for lignin depolymerization can be performed by fungal as bio-catalytic to lignin degradation. This process can be also done by certain bacteria or consortia bacteria (Cho et al., 2004). Jääskeläinen et al. (2003) found a high yield (83%) of lignin by enzymatic hydrolysis.

On the other hand, there are a variety of methods that can be used to convert bagasse into non energy products, such as physical methods, chemical methods, mechanical extraction methods and biological methods. The selection of the adequate process of treatment depends on the final goal of treatment.

The physical and chemical pretreatment methods may be used to break down biomass components. The physical pretreatment refers to reduce the size of raw material by mechanical actions to make it accessible into successive treatment such as biological and/or chemical treatment.

The major techniques employed to break down biomass components are based on the hydrolysis method. This method, which is sometimes called saccharification, that is used to break down the hydrogen bonds in the lignocellulosic biomass fractions by removal of the surrounding hemicellulose and/or lignin along with modification of the cellulose microfibril structure and recovering the resulting soluble monomeric and/or oligomeric sugars (Motaung and Anandjiwala, 2015; Rocha et al., 2015b).

The steam explosion is a physical pretreatment method that is used for cellulosic materials pretreatment to the solubilization and separation of one or more of the four major components of biomass (hemicellulose, cellulose, lignin, and extractives). During this process, steam permeates the biomass material and initiates a chemical autohydrolysis reaction, because some organic acids that are presented in biomass can be catalyzed and help remove of the hemicellulose and lignin from their biomass structure (Pol et al., 2015). This phenomenon is observed during the steam processes, such as the formation of acetyl groups (Rocha et al., 2015a,b). Sometimes, chemicals, such as sulfuric acid, can be added to improve efficiency of the process (Rocha et al., 2015b). On the other hand, the most common chemical pretreatment methods make use of dilute acid, alkaline, organic solvent, ammonia, sulphur dioxide, carbon dioxide or other chemicals to make the biomass more adequate for further applications (Motaung and Anandjiwala, 2015).

Rocha et al. (2015a), in their investigation of 60 sugarcane samples, presented ratios of mass fractions of lignin extracted from sugarcane bagasse which are determined by pretreatment of concentration sulfuric acid method. These ratios are 7.4/1 (total lignin/soluble lignin), 1.2/1 (total lignin/insoluble lignin) and 0.2/1 (soluble lignin/insoluble lignin). This means that the extraction of lignin need successive treatment, because the major of lignin is insoluble. On the other hand, the acid method is not adequate to treat lignin, but it can treat hemicellulose effectively (Lavarack et al., 2002; Fogel et al., 2005; Geddes et al., 2010; Rocha et al., 2011).

The alkaline hydrolysis process is a breakdown of the intermolecular ester bonds cross-linking with lignin and other material components as celluloses and hemicelluloses. It is more effective to solubilize a greater fraction of lignin (Mosier et al., 2005; Rocha et al., 2012). Rocha et al. (2012) demonstrated that the steam explosion and alkaline delignification processes are good for extraction of lignin from bagasse.

3. Extraction and isolation of lignin from sugarcane bagasse

Establishing and selecting a method for isolation and recovery of lignin from non-wood biomass such as bagasse is not an easy task. However, there are several methods that are recently developed to extract and isolate the lignin from sugarcane bagasse, such as alkaline method (Li and Ge, 2011; Moubarik et al., 2013; Wahba et al., 2015), and ionic liquid (Tan and MacFarlane, 2009; Tan et al., 2009).

3.1. Alkaline methods

In this section, a briefing of some extraction and isolation of solubilized lignin from sugarcane bagasse methods, is presented (Wahba et al., 2015; Moubarik et al., 2013).

3.1.1. Alkaline pulping methods

The delignification method is based on the treatment by soda solution (alkaline delignification), usually after pre-treatment by mechanical and hot water, at about 100 °C for about one hour or more, with different concentrations of soda and biomass (solid) to liquid ratio; briefly:

3.1.1.1. First step (Pre-treatment). Sugarcane bagasse was air-dried at ambient temperature for three days until humidity reach to 8–10%, then cut into small pieces (1–2 cm), grounded and screened to pass a through 1.0 mm size screen. The dry powder is treated with hot water at 70–80 °C for 2 h with solid-to-liquid ratio 1:10 w/w, and then it is cooled to 25 °C, washed with water at the same above mentioned ratio and centrifuged for 10 min at 2000 rpm to separate the solubilized hemicelluloses.

3.1.1.2. Second step (Pulping method). The separated solid was pulped at 90–98 °C by sodium hydroxide solution (15% NaOH m/v) with solid to liquid ratio 1:10–1:18 w/w at 90 °C for 1.5–4 h, under constant stirring, then filtered to obtain the black liquor without any fibrous materials. The filtrated (black liquor) was acidified by sulfuric acid (5N H₂SO₄) to pH 2, in order to precipitate the acidified lignin then washed by water and collected by centrifugation and air dried, schematically illustrated in Fig. 2 (Wahba et al., 2015; Moubarik et al., 2013).

The yield of lignin obtained from this method was 13%. The two sequential stages of treatment enabled to isolate 86% of original lignin from sugarcane bagasse.

3.1.2. Other similar methods

3.1.2.1. Soda-pulping method. Costa et al. (2017) used the lignin extracted from sugarcane bagasse for assembling micro-particles. The extraction and modification of lignin derivative from sugarcane bagasse done by soda-pulping process to solubilize the lignin present in biomass, as following:

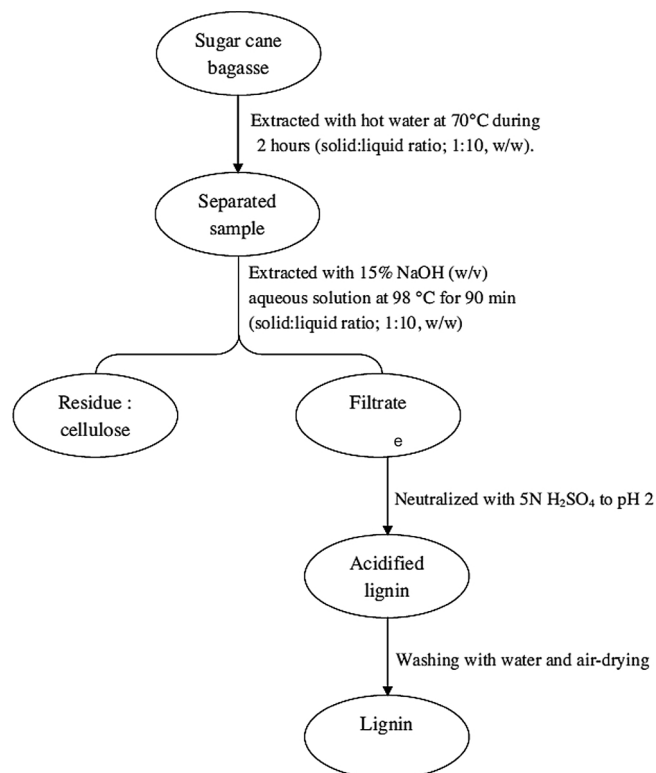


Fig. 2. Illustration method of extraction and isolation of lignin from sugar cane bagasse (adapted from Moubarik et al., 2013).

- First step, the bagasse is washed with hot water at 80 °C, then it is dried for 4 days at 40 °C in a circulating air oven until 10% w/w moisture content is obtained, and then it is milled;
- The milled powder was immersed in aqueous solution, 0.5 mol L⁻¹ NaOH (1:10 m/v) with ratio 1:10 g/mL, then autoclaved for 1 h at 120 °C and then filtered. The pH of filtrated (black liquor) was adjusted to 5.5, with 1.0 mol L⁻¹ of HCl, and then concentrated to obtain a small volume (reduced);
- Adding three volumes of 95% ethanol (v/v) to the resulting liquor for precipitation hemicellulose and silicon compounds. After this, the ethanol present in the resulting liquor is removed and the pH is lowered to 1.5 for precipitation of lignin, the lignin obtained was filtered, washed, and dried (lignin obtained with this step is so called Lig-Alk).
- The second type of lignin was obtained from the black liquor (200 mL) and used directly in the oxidation process by adding 14% H₂O₂ (w/v) to reflux at pH 13, for 2 h at 98 °C, then purified and precipitated as described above (lignin obtained with this step is so called Lig-Ox).
- Other type of lignin was obtained through the acetylation of Lig-Alk by dissolving 1.0 g of Lig-Alk in 10.0 mL of anhydrous pyridine and 20.0 mL of anhydrous acetic anhydride under an inert atmosphere and agitated at 28 °C for 2 days, then added 20.0 mL of dichloromethane/MeOH solution (1:1 v/v). The resulting solution separated with 2.0 mol L⁻¹ of HCl until pyridine had been completely removed, then washed with 10.0 mL of water and dried (lignin obtained with this step is called Lig-Ac).

The total yield of precipitated lignin from this method according to sugarcane bagasse weight are: Lig-Alk: 7.2%; Lig-Ox: 5.5%; and Lig-Ac: 6.2%.

3.1.3. Extraction by dioxane and hydrochloric acid and by alkali solution

Hoareau et al. (2004), studied the bagasse lignin that pretreated by cyclohexane-ethanol. The methods of extraction and modification of lignin from sugarcane bagasse are as following:

3.1.3.1. Pretreatment. The extraction of sugarcane bagasse was done in Soxhlet for 2 days with a mixture of cyclohexane-ethanol (1:1, v/v) then with water for one day and dried until constant weight in an air-circulated stove at 60 °C, then grounded to obtain a powder (1 mm diameter).

3.1.3.2. Treatment by dioxane and hydrochloric acid. The dry powder treated with a dioxane solution and 0.1 N aqueous hydrochloric acid (8.5:1.5, v/v) with ratio (100 g/1L) and heated under nitrogen at 100 °C for 2 h, then filtered and washed three times with a fresh dioxane-water solution (3 × 500 mL, 8.5:1.5) without hydrochloric acid. The filtrates concentrated by evaporation under vacuum to eliminate dioxane in conditions that maintain the pH solution above 1 to avoid excessive depolymerisation of lignin, then centrifuged and the precipitation dried under vacuum (30 mbar) over phosphorous anhydride as purity of bagasse lignin. The purity of bagasse lignin is 90% related to soluble lignin and the yield of acidolysis lignins isolated from bagasse is estimated about 10% of fiber treated. (Hoareau et al., 2004).

3.1.3.3. Treatment by alkali (KOH) solution. Sugarcane bagasse (dry powder) stirred with a 0.5 M aqueous KOH solution with ratio 50 g/1.5L under nitrogen at 35 °C for 3.5 h, then filtered and washed with water then with ethanol. The filtrate is neutralized with a 6 M acetic acid solution to pH 5.5.

This filtrate is treated by three volumes of 95% ethanol for precipitation of hemicellulose fraction (hemicellulose isolation). Then after evaporation of ethanol, the concentrated filtrate is precipitated in water and adjusted with a 6 M HCl solution to pH 1.5 to obtain alkali lignin (lignin isolation). Then centrifuged and washed with acidified water to

pH 2.0 and then dried under vacuum over phosphorous anhydride. The purity of bagasse lignin obtained with this method is 95% related to soluble lignin and the yield of acidolysis lignins isolated from bagasse is estimated to be about 8% of biomass treated (Hoareau et al., 2004).

3.1.4. Sequential isolation steps by alkaline dioxane

Zhang et al. (2013) studied the extraction of lignin fractions derived from ball-milled sugarcane bagasse by alkaline dioxane extraction processes. Three steps of extractions, successively, were done using 96% dioxane, 50% dioxane, and 80% dioxane containing 1% NaOH at boiling temperature.

3.1.4.1. First step extraction with 96% dioxane. Bagasse treated with dioxane solution (96:4, v/v dioxane/water) with ratio 1:20 (g bagasse/mL solution) for two hours, then filtrated and washed with same solution until the filtrate was clear. The filtrates were concentrated to about 50 mL then transferred into 3 vols with 95% ethanol then the precipitates of this operation were filtered and washed with 70% ethanol and freeze-dried to obtain hemicellulosic fraction (96% dioxane-soluble); this fraction is called H₁. The filtrates were concentrated to remove ethanol and resting dioxane then transferred into 10 vols of H₂O adjusted with 6 M HCl (pH 1.5–2.0). Then, the precipitates were filtered and washed with acidified water (pH 2.0), and freeze-dried to obtain the lignin fraction (96% dioxane-soluble), this fraction is called L₁.

3.1.4.2. Second step extraction with 50% dioxane. The solid residues obtained after first treatment (R₁) dried for 16 h at 50 °C in a cabinet oven with air circulation, then, successively, treated with dioxane solution (50:50, v/v dioxane/water) with same ratio and time in the first treatment, then filtrated and washed with same solution until the filtrate was clear. The filtrates were treated using the same method described above to obtain hemicellulosic fraction (50% dioxane-soluble), which is called H₂, and the lignin fraction (50% dioxane-soluble), which is called L₂.

3.1.4.3. Third step extraction with 80% dioxane. The solid residues obtained after second treatment (R₂) were refluxed with dioxane solution (80:20, v/v dioxane/water) containing 1% NaOH at boiling temperature with same ratio and time described above, then filtrated and washed with same solution used in this treatment until the filtrate was clear. The filtrates were adjusted to pH 5.5–6.0 and treated using the same methods described above to obtain hemicellulosic fraction (80% dioxane-soluble), this fraction is called H₃, and lignin fraction (80% dioxane-soluble), is called L₃, (Zhang et al., 2013).

The sequential isolation steps of lignin from sugarcane bagasse with 96% dioxane, 50% dioxane, and 80% dioxane containing 1% NaOH at boiling temperature were extracted 5.10%, 1.35% and 0.75%, respectively. The total yields of lignin were 7.2% based on dried sugarcane bagasse. In the first step, 70.8% of the total extracted lignin was isolated.

3.1.5. Other sequential methods

Sun and his co-workers (Sun et al., 2003) presented eight fractions of lignin that have been isolated from sugarcane bagasse as following:

- 1) **Milling:** Sugarcane bagasse first dried in sunlight, milled and screened to pass through an 0.8 mm sieve.
- 2) **Dewaxed:** 10 g of dried powder extracted in a Soxhlet apparatus by toluene/ethanol (2:1, v/v) for 6 h, then dried in an oven at 60 °C for 16 h.
- 3) **Distilled water:** then the obtained powder from phase 2) is immersed in 300 mL of distilled water at 55 °C for 2 h under stirring, then treated by 3 vols of 95% ethanol to precipitated water extract (water-soluble hemicellulose). The supernatant solution acidified by 6 M HCl to pH 1.5 to obtain soluble lignin fraction L₁.

- 4) **0.5 M NaOH:** The residues free of wax and water soluble successively treated with 300 mL of 0.5 M NaOH then filtrated and washed with distilled water until the pH of the filtrate was neutral, and then dried at 60 °C. The supernatant fluids adjusted to pH 5.5 with 6 M HCl and then concentrated to about 30 mL under reduced pressure, and then carried into 120 mL by 95% ethanol to precipitated hemicellulose. The solubilized lignin obtained from the corresponding supernatant by precipitation at pH 1.5–2.0 (lignin fraction L₂).
- 5) **0.5% H₂O₂:** The residue from phase 4) treated by 200 mL of 0.5% H₂O₂ at pH 11.5 adjusted with 6 M NaOH then repeated the procedures in phase 4) to obtain soluble lignin fraction L₃.
- 6) **1.0, 1.5, 2.0, and 3.0% H₂O₂:** Sequential treatment: same procedures in phase 5 repeated with various concentrations of Alkaline Peroxide (1.0, 1.5, 2.0, and 3.0% H₂O₂) to obtain lignin fraction L₄, L₅, L₆ and L₇.
- 7) **2M NaOH:** the last residue is treated by 200 mL of 2 M NaOH at 55 °C for 2 h then filtrated and washed with distilled water until the pH of the filtrate was neutral, and then dried at 60 °C. The supernatant fluids adjusted to pH 5.5 with 6 M HCl and then concentrated to about 30 mL under reduced pressure, then carried into 120 mL by 95% ethanol to precipitated hemicellulose. The solubilized lignin obtained from the corresponding supernatant by precipitation at pH 1.5–2.0 (lignin fraction L₈).

The acid-insoluble lignin fractions obtained from each of the above lignin fractions after rinsed by acidified water at pH 2.0 are freeze-dried.

The total extracted of lignin from the sequential isolation steps was 16.1% on dry based.

Table 1 shows the percentage of lignin isolated from sugarcane bagasse by different methods. The sequential treatments with various concentrations of alkaline peroxide and sodium hydroxide released high quantity of lignin. This indicates that these sequential treatments of bagasse with alkali and alkaline peroxide are suitable to isolate lignin from sugarcane bagasse. The method of alkaline pulping with sodium hydroxide solution is one of the most suitable methods for isolation of lignin from sugarcane bagasse.

3.2. Ionic liquid methods

Recently, ionic liquids (as: 1-ethyl-3-methylimidazolium acetate ([EMIM]OAc); 1-allylmethylimidazolium chloride ([AMIM]Cl); 1-butyl-3-methylimidazolium chloride ([BMIM]Cl)) used as a solution for pre-treatment, fractionating and removing lignin from lignocellulose biomass (Espinoza-Acosta et al., 2014).

Lan et al. (2011), used the Ionic Liquid (IL) method for fractionation of bagasse into cellulose, hemicelluloses, and lignin. This method based on dissolution of bagasse in 1-butyl-3-methylimidazolium chloride ([C4mim]Cl) followed by precipitation in acetone/water (9:1, v/v) and extraction with 3% NaOH solution. This could be briefly described as following:

3.2.1. Pretreatment

Sugarcane bagasse dried, in sunlight, and then cut into small pieces, ground and screened to pass throw 40–60 mesh and then dried, in a cabinet oven with air circulation, for 16 h at 55 °C. This dried extracted with toluene/ethanol (2:1, v/v) and further pulverized with a vibratory ball mill for 72 h.

3.2.2. Treatment

The ball milled bagasse dissolved in [C4mim][Cl], with ratio 1:50 (w/w), at 110 °C under inert environment with agitation to obtain a clear solution, and then add into 200 mL of acetone/water (9:1, v/v) under strong agitation. The residue 1 (Fig. 3) separated by centrifugation at 4000 rpm for 15 min, collected and washed by acetone/water

Table 1
Comparison of various lignin isolation methods from sugarcane bagasse.

Methods	Treatment conditions	Lignin isolated (% dry matter)	References
Alkaline pulping with sodium hydroxide solution	Hot water pretreatment with ratio 1:10 S/L, at 70–80 °C for 2 h.	13.0	Wahba et al. (2015); Moubarik et al. (2013)
Alkaline pulping with sodium hydroxide solution	Pulping with 15% NaOH (m/v) and ratio 1:10–1:18 (w/w) at 90–98 °C for 1.5–4 h.	7.2;	Costa et al. (2017)
	Hot water pretreatment at 80 °C. Pulping with NaOH (1:10 m/v) and ratio 1:10 (g/mL) at 120 °C for 1 h,	5.5;	
	a) Adding three volumes of 95% ethanol (v/v) and acidified until pH to 1.5 → lignin obtained referred as Lig-Alk b) oxidation by 14% H ₂ O ₂ (m/v) at pH 13, for 2 h at 98 °C → lignin obtained referred as Lig-Ox c) acetylation of Lig-Alk under an inert atmosphere and agitating at 28 °C for 2 days → lignin obtained referred as Lig-Ac	6.2	
Extraction by dioxane and hydrochloric acid and by alkali solution	Pretreatment by cyclohexane-ethanol (1:1, v/v) for one day and dried at 60 °C,	10.0	Hoareau et al. (2004)
Alkaline dioxane extraction	A) Treatment by dioxane solution and (0.1 N) aqueous hydrochloric acid (8.5:1.5, v:v) with ratio (100 g/L) and heated under nitrogen at 100 °C for 2 h Treatment by alkali (0.5 M KOH) solution with ratio 50 g/1.5 L under nitrogen at 35 °C for 3.5 h	8.0	Zhang et al. (2013)
	Sequential steps: a) Treatment with dioxane solution (96:4, v/v dioxane/water), ratio 1:20 (g bagasse/mL solution) for 2 h.	5.1; 1.4;	
	b) Treatment with 50% dioxane (50:50, v/v dioxane/water), ratio 1:20 (g bagasse/mL solution) for 2 h. c) Treatment with 80% dioxane (80:20, v/v dioxane/water) containing 1% NaOH at boiling temperature, ratio 1:20 (g bagasse/mL solution) for 2 h.	0.8	
Sequential treatments with various concentrations of alkaline peroxide and sodium hydroxide	Sequential treatments steps with distilled water, 0.5 M NaOH, 0.5, 1.0, 1.5, 2.0, and 3.0% H ₂ O ₂ at pH 11.5, and 2 M NaOH at 55 °C for 2 h	0.5, 9.5, 2.7, 0.6, 1.0, 0.9, 0.5, and 0.4	Sun et al. (2003)

(9:1, v/v) to eliminate [C4mim]Cl, and then dried and extracted with 3% NaOH solution at 50 °C with ratio 1:25 (g/mL) for 45 min.

The filtrate 1 collected and concentrated by evaporation to remove acetone and then added into 250 mL of acidified water (pH 2.0) under stirring. The residue 2 centrifuged and washed with acidified water (pH 2.0) to obtain acetone soluble lignin. Filtrate 2 reused after treated as recycled IL (Fig. 3).

The cellulose obtained (residue 3) from residue 1 after treatment by 3% sodium hydroxide solution was washed three times with distilled water. The filtrate 3 was treated with 4 M hydrochloric acid (HCl) and precipitated in 3 vols using 95% ethanol to obtain hemicellulose (residue 4). The filtrate 4 was concentrated under reduced pressure to remove ethanol and precipitated in acidified water (pH 2.0) to obtain alkaline lignin (residue 5).

The fractionations of bagasse obtained from this method are 36.78% cellulose, 26.04% hemicelluloses, 10.51% lignin and 26.67% of biomass lost.

The following Table (Table 2) shows the percentage of lignin removed from sugarcane bagasse by different ionic liquid and treatment conditions. The 1-(4-sulfobutyl)-3-methylimidazolium hydrosulfate is the best method that could be used to remove all lignin. The 1-Ethyl-3-methylimidazolium sodium alkylbenzenesulfonate ([EMIM][Na][ABS]) is the second best method that removed 93%.

4. Composition and structure of sugarcane bagasse

The chart in Fig. 4 shows the chemical compositions of sugarcane bagasse that can be separated from one or more of its major components, such as cellulose, hemicellulose, and lignin. This biomass material is a natural complex biopolymer structure of carbohydrates, chemically has four important types of bonds between the monomer units: ester bonds, ether bonds, hydrogen bonds and carbon-to-carbon bonds (Harmsen et al., 2010). Ester bonds are between lignin and polysaccharides (cellulose and hemicellulose), ether bond are among the functional groups (as hydroxyl group), hydrogen bonds that is between the cellulose polymer chains, and carbon-to-carbon bonds (as in

aromatic ring). Functional groups related to lignin in components of lignocellulose are aromatic ring, hydroxyl group; carbon to carbon linkage and ether (glucosidic) linkage.

Cellulose fraction, (C₆H₁₀O₅)_n, is a long-chain polymer polysaccharide, sometimes so-called “glycans” carbohydrate of beta-glucose, which forms the primary structural component of plants. Hemicellulose fraction, (C₅H₈O₄)_n, is similar to cellulose but is less complex; it consists mainly of pentosic sugars and sugar acids.

Lignin (*lignum*) is a complex polymer, [(C₉H₁₀O₃(OCH₃)_{0.9–1.7})_n], combines cellulose and hemicellulose (Akhtar et al., 2015).

Many researchers have established that the lignin derivative from non-wood biomass is made up of three principal monolignols units, these polymers are derived from the *p*-hydroxycinnamyl alcohol monomers. Three principal monolignols subunits are *p*-Hydroxyphenyl subunit (H), Guaiacyl subunit (G), and Syringyl subunit (S). They are linked between them through different types of linkages. They are *p*-coumaryl alcohol (C₉H₁₀O₂), coniferyl alcohol (C₁₀H₁₂O₃), and sinapyl alcohol (C₁₁H₁₄O₄), or *trans*-*p*-coumaryl, *trans*-coniferyl, and *trans*-sinapyl alcohols (Sun et al., 2003; Arni et al., 2007; Arni et al., 2010a,b; Lino et al., 2015). Due to the presence of H lignin in addition to G and S, lignin in bagasse differs from those in hardwood (Brandt et al., 2013).

Bagasse is an excellent resource material for production of chemical compounds (Fig. 4), such as arabinose, glucose, furfural (Lavarack et al., 2002), lactic acid (Laopaiboon et al., 2010), xylose (Boussarsar et al., 2009), and polymetric materials such as plastics (Doherty et al., 2007).

The lignin fraction contains phenolic acids and aldehydes which have chemical compounds such as *p*-hydroxybenzoic acid, *p*-hydroxybenzaldehyde, vanillic acid, vanillin, syringic acid, syringaldehyde, acetovanillone, acetosyringone, *p*-coumaric acid, and ferulic acid (Sun et al., 2003; Arni et al., 2007; Arni et al., 2010a,b). In addition, the sugar of lignin fraction has rhamnose, arabinose, xylose, mannose, galactose and glucose (Sun et al., 2003; Arni et al., 2007).

Lignin is rich with natural aromatic organic compounds such as *p*-coumaric acid (*p*-CA), ferulic acid (FA), syringic acid (SA), vanillin (V) (Arni et al., 2007; Arni et al., 2010a,b).

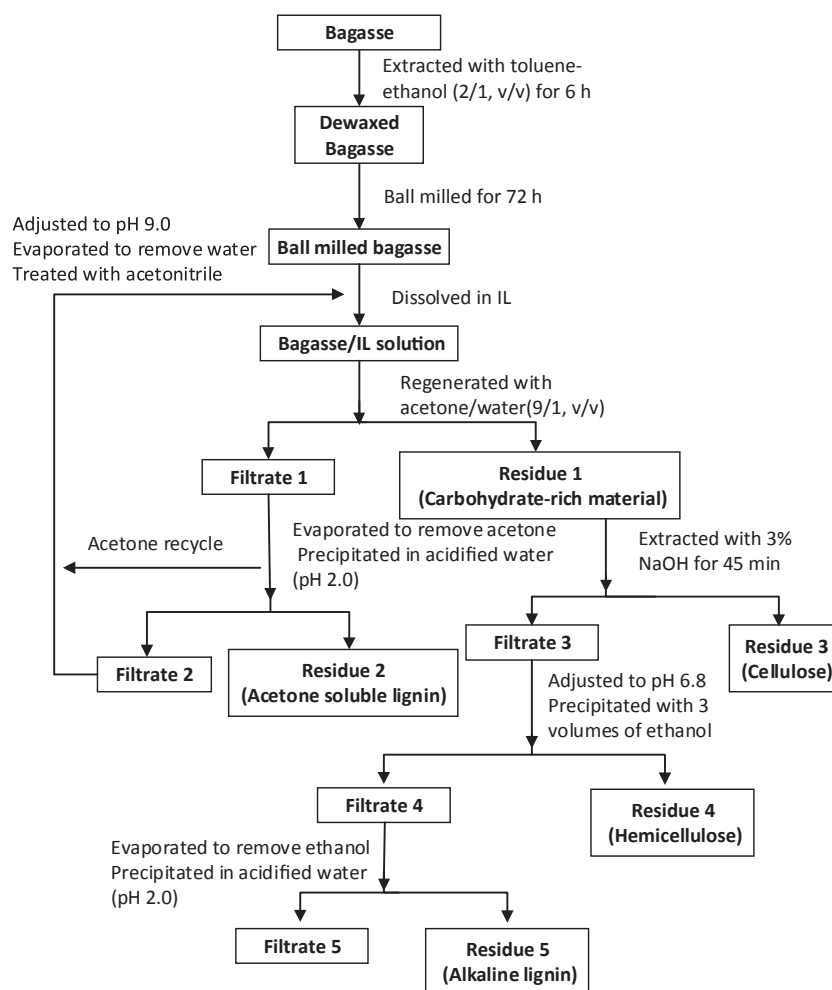


Fig. 3. Schematic illustration of Ionic Liquid method (adapted from Lan et al., 2011).

5. Quantity of sugarcane lignin fraction

The quantity of sugarcane lignin fraction depends on the method of extraction. The determination of lignin content in biomass is usually determined by the Klason method. The Klason method or treatment by concentrated sulphuric acid, usually 72%, is used to separate the insoluble lignin from lignocellulosic materials. The fraction of lignin separated by this method is called “Klason lignin”. The quantity of sugarcane lignin fraction (Klason lignin) is in the range of 17–32% of lignocellulosic bagasse on dry weight of biomass (Table 3). Table 3 shows the range of lignocellulosic materials. It provides a comparison

between sugar cane bagasse wood and non-wood materials.

Table 4 shows the fractions of sugarcane bagasse collected from literature and the chart (Fig. 4) shows its chemical composition based on the mass balance.

On the other hand, the percentage of lignin fraction does not highly depend on the cultivation places. The range that above mentioned which treated by H_2SO_4 72%, it has 22% as an average over 60 samples collected from different places (Rocha et al., 2015a).

Other methods are based on the soluble lignin, which could be extracted and isolated using organic solvents (such as acetic acid, ethanol, dioxane, methanol, etc.), ionic liquids (IL) and alkaline solutions. The

Table 2

Comparison of ionic liquid methods used to remove lignin from sugarcane bagasse.

Ionic liquid	Treatment conditions	Lignin removed (%)	References
1-Ethyl-3-methylimidazolium sodium alkylbenzenesulfonate ([EMIM][Na][ABS])	170–190 °C, 30–120 min	93.00	Tan et al. (2009)
1-butyl-3-methylimidazolium chloride ([C4mim]Cl)	110 °C, 15 min under a nitrogen atmosphere with agitation	54.62	Lan et al. (2011)
1,3-dimethylimidazolium dimethyl phosphate ([Mmim][DMP])	120 °C, 120 min	n.d.	Bahrani et al. (2015)
1-Ethyl-3-methylimidazolium acetate ([EMIM][OAc])	165 °C, 10 min	34.00	Li et al. (2011)
[EMIM][OAc]	120 °C, 30 min	32.00	Qiu et al. (2012)
1-butyl-3-methylimidazolium methyl sulfate ([BMIM][MeSO ₄]) (7% H ₂ SO ₄)	125 °C, 2 h	26.00	Diedericks et al. (2012)
1-(4-sulfobutyl)-3-methylimidazolium hydrosulfate ([BMIM][SO ₃][HSO ₄]) (ethanol/water)	200 °C, 30 min	100.00	Long et al. (2013)

n.d. = not determined.

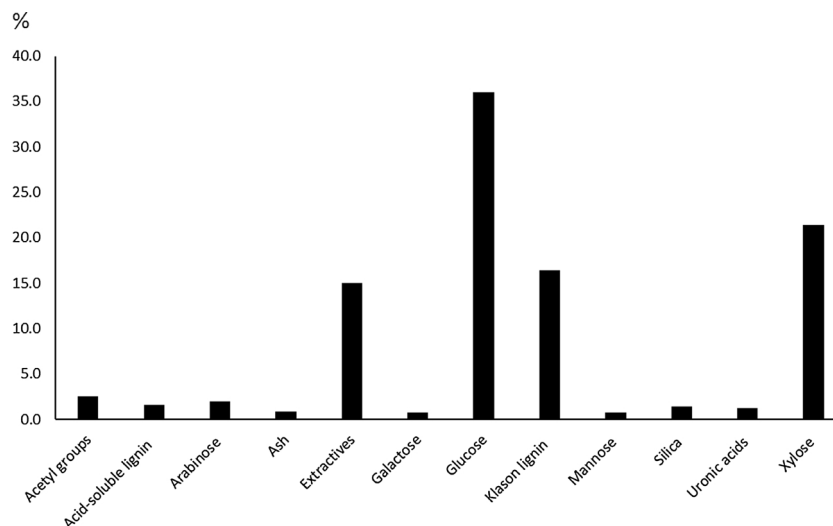


Fig. 4. Chemical composition of bagasse based on the mass balance (adapted from Carvalho, 2015).

solubilized lignin is then recovered from solution by precipitation with acids (Mancera et al., 2010). The solvents that solubilizes the lignin from biomass, in different quantities, depend on the applied techniques and the aim of study. The following are some examples of the quantities extracted and isolated by these procedures:

- The range of percentage of lignin fraction on dry weight is between 9%, for that treated by NaOH 4%, and 30% for those treated by H₂SO₄ 1% (Rezende et al., 2011).
- On the other hand, the total yield of lignin is 7.2% based on dried biomass treated by alkaline dioxane extraction processes. The 96% dioxane process gives a 70.8% of the total extracted lignin that it was isolated (Zhang et al., 2013).
- The yields of lignin that be treated by dioxane and hydrochloric acid and by alkali (KOH) solution are 10% and 8%, respectively, reference to Klason lignin (Hoareau et al., 2004).
- The total yield solubilized lignin that has been isolated from the sugarcane by sequentially sodium hydroxide and alkaline peroxide is 89.0% of the original lignin bagasse (Sun et al., 2003).

The average value of lignin fraction is about 23% of biomass (Table 4), this value is similar to that found by Rocha et al. (2015a) and it is independent of the cultivation zones.

6. Comparison between lignin fractions in sugarcane bagasse and in non-wood biomass

Generally, non-woods contain less lignin, but the percentage of Klason Lignin of bagasse has a major content among non-wood biomass

Table 3
Composition range of lignocellulosic materials.

Values are in percentage					
Biomass species	Cellulose	Hemicellulose	Lignin	Extractive	Ash
Hardwood	40.0–55.0	24.0–40.0	16.0–25.0	2.0–8.0	0.2–0.6
Softwood	40.0–50.0	24.0–35.0	25.0–35.0	1.0–5.0	0.2–0.6
Sugarcane bagasse	32.0–45.0	20.0–32.0	17.0–32.0	1.0–9.0	1.0–3.0
Non-wood biomass	29.2–73.8	9.9–38.8	7.5–26.1	0.8–8.3	0.9–14.0

Table 4
Fractions of sugarcane bagasse collected from literature.

Values are in percentage, w/w, on dry weight					
References	Cellulose	Hemicellulose	Lignin	Extractive ¹	Ash
Rocha et al. (2015a)	42.19 ^a	27.60 ^a	21.56 ^a	5.63 ^a	2.84 ^a
Canilha et al. (2011)	45.00	25.80	19.10	9.10	1.00
Sun et al. (2003)	43.60	33.50	18.10	0.80	2.30
Hoareau et al. (2004)	55.20	16.80	25.30	n.d.	1.10
Ezhumalai and Thangavelu (2010)	50.00	25.00	25.00	n.d.	n.d.
Yao et al. (2015)	45.28	22.13	22.39	7.11 ^b	1.01
Pereira et al. (2011)	45.40	28.70	23.40	n.d.	2.70
Rabelo et al. (2011)	40.50	24.50	26.40	7.00 ^c	1.60
Ayeni et al. (2015)	35.28	33.28	25.20	2.14	4.10
Average value	44.72	26.37	22.94	3.53	1.85

¹ Mainly of the extractive is wax.

^a Average of 60 samples.

^b Benzene–alcohol extract (%) [the 1% Sodium hydroxide extract (%) is 24.97].
n.d. = not determined.

(Kirk-Othmer, 2005); it has an average of 23% of biomass compared to 16.2% of non-wood (Table 5). This comparison of sugarcane bagasse with various non-wood plants reveals that this material has the highest content of lignin and a lowest content of ash than all non-wood biomass (Table 5).

The highest amount of lignin in bagasse could be considered as an economic advantage towards deriving chemical resources from agro-industrial waste residues.

Table 5

The composition of some non-wood biomass collected from literature.

Biomass	Cellulose	Hemicellulose	Lignin	Extractive	Ash	References
Values are in percentage						
<i>Arundo donax</i> (Giant reed: <i>Arundo donax</i> L.)	29.20	32.10	20.90	n.d.	4.80	Shatalov et al. (2001)
Banana (<i>Musa paradisiaca</i>) pseudo-stems	40.00	25.20	12.70	n.d.	14.00	Cordeiro et al. (2004)
Amaranth (<i>Amaranthus</i>)	32.00	26.10	13.20	n.d.	12.00	Fiserova et al. (2006)
Orache (<i>Atriplex</i>)	36.00	38.80	19.50	n.d.	2.00	Fiserova et al. (2006)
Curauá (<i>Ananas lucidus</i>)	73.60	9.90	7.50	n.d.	0.90	Hoareau et al. (2004)
<i>Ferula orientalis</i> L.	45.60	22.57	26.11	0.87	4.85	Aysu and Küçük (2014)
Wheat (<i>Triticum aestivum</i>) straw	30.60	32.30	16.80	9.80	5.00	Silva-Fernandes et al. (2015)
Bamboo (<i>Bambusa vulgaris</i>) fiber	73.83	12.49	10.15	3.16	n.d.	Abdul Khalil et al. (2012)
Oil palm (<i>Elaeis guineensis</i>) empty-fruit-bunch (OPEFB)	62.90	28.00	18.80	3.70	1.30	Law et al. (2007)
Barley (<i>Hordeum vulgare</i>) straw	36.20	28.30	16.10	8.30	9.32	Ferrer et al. (2015)
Average value	46.00	25.60	16.20	2.60	5.40	

n.d. = not determined.

7. Conclusion

Bagasse is a potentially reliable and excellent renewable raw resource material for several applications, starting from domestic to commercial use. Two categories of processing technologies can be used for the conversion of bagasse into energy and not into energy products. Several physical and chemical methods may be applied for extraction and isolation of lignin from sugarcane bagasse, such as alkaline and ionic liquid methods. Lignin fraction has an average of about 23% of biomass, this value is a highest compared to various non-wood biomass wastes. It is abundant with chemical compounds, such as aromatic compounds.

It can be concluded that lignin fraction is economically and environmentally advantageous for deriving chemical resources from agro-industrial waste residues.

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